Experiment - 4

18/08/2026 Setup & configure a LAN using a Switch & Ethernet cables in your lab

Aim :- To design, set up & configure a local Area Network (LAN) connecting four computers through a network switch & demonstrate resource/data sharing across the network

Equipment Required :-

Equipment	Quantity	Specification
Personal Computer	4	Ethernet (NIC) enabled
Network Switch	1	8 / 16 / 24 port
Ethernet Cables (Cat 5e / Cat 6)	4	Straight through RI-45 connectors

Physical x: 912, y: 387

PC-PT
PC0

PC-PT
PC1

2960-24T
Switch0

Server-PT
Server0

PC-PT
PC2

PC-PT
PC3

PC-PT
PC12

Hub-PT
Hub1

PC-PT
PC13

PC-PT
PC14

PC-PT
PC8

PC-PT
PC9

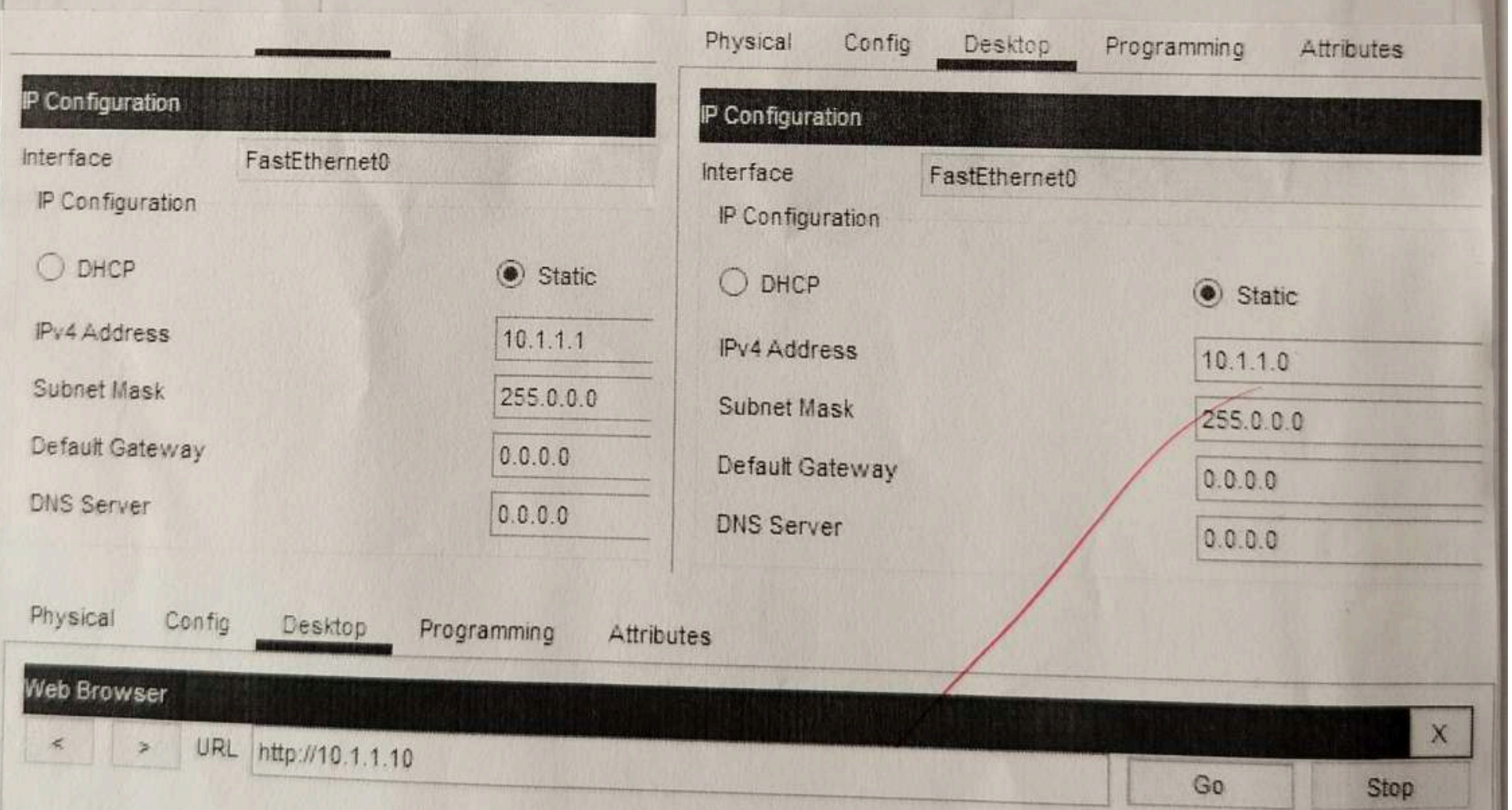
2960-24T
Switch1

PC-PT
PC10

PC-PT
PC11

IP Address Configuration Table :-

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC_2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC_3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch	10.1.1.5	255.0.0.0	WLAN Interface



Implementation in CISCO Packet Tracer

1) Place the Devices

* Open CISCO packet tracer.
* From the device category panel, select End Devices & drag 4 PC onto the workplace.
* Select switches & drag switch.

2) Cable the devices

* Click Connections → Select Copper Straight - through cable
* Click each PC, then click the Switch & select an available Fast Ethernet port to complete each connection.
* Repeat for all 4 PC's (ports Fa 0/1 to Fa 0/4) Green link lights confirm a successful physical connection.

3) Assign IP Addresses to PC's

* Click a PC → Desktop Tab → IP Configuration
* Enable the Static IP Address & Subnet Mask as per the IP Table.

④ Configure the switch management IP

* Click the Switch → CLI Tab
* enable
  configure terminal
  interface Vlan1
  ip address 10.1.1.5 255.0.0.0
  no shutdown
  exit

⑤ Verify Connectivity using Ping

* Click a PC → Desktop Tab → and prompt
* Use the ping to test connectivity
* Successful message

⑥ Demonstrate Data Sharing (Server Simulation)

* Add a server device (End devices → Server-PT) & connect it to the switch.
* Assign it an IP address.
* Open the server's services tab & enable the HTTP.
* From another PC, open Web Browser & enter the server's IP address to load the page confirmation successful data transfer across the LAN.

Result : This LAN is settled up & connection is verified successfully.

28/8/25